

Business Process Mining

CoatXR P2P Process Analysis

Business Process Mining Project Report

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Team Members	Tasks Division
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Task 1 Event Log Inspection

Procedure

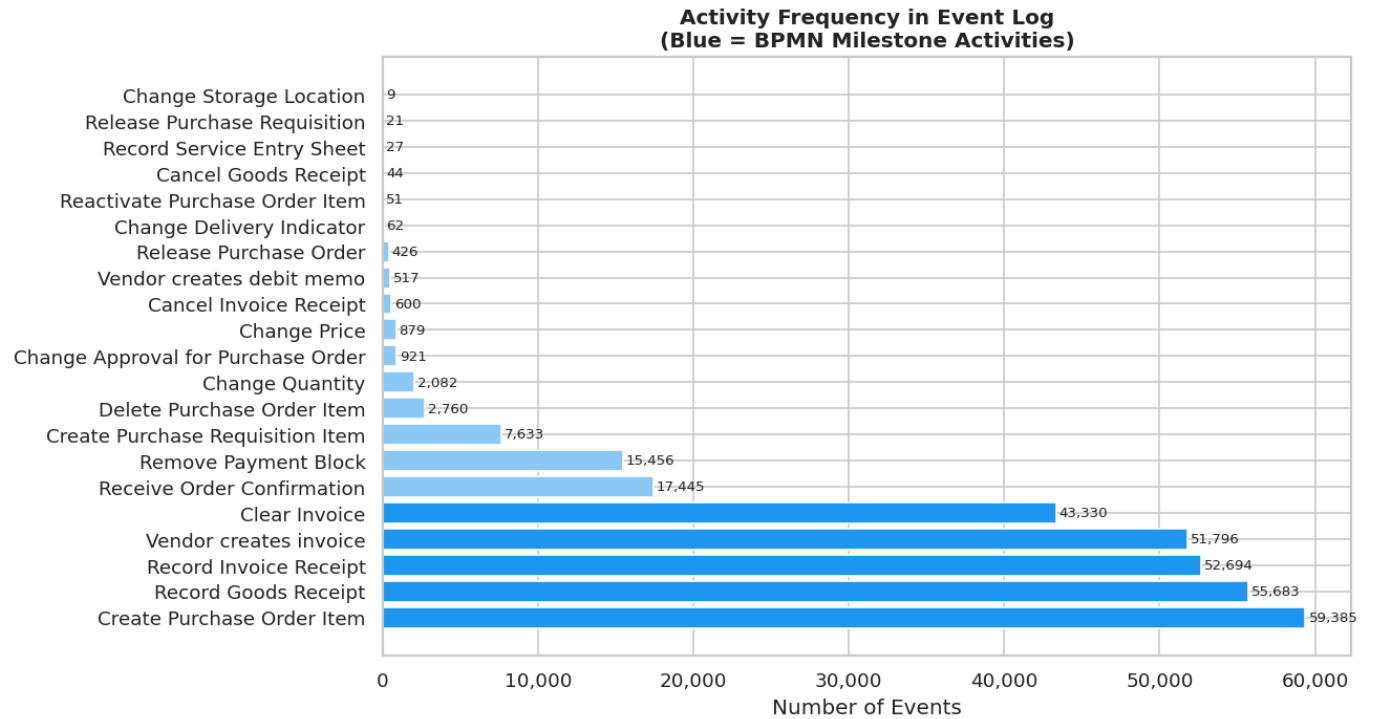
The event log was inspected using Python (pandas, matplotlib) in Google Colab. All 22 columns were analysed for data type, unique value counts, null rates, and relationships. Cases were classified as Completed or Ongoing per the project definition.

Key Figures

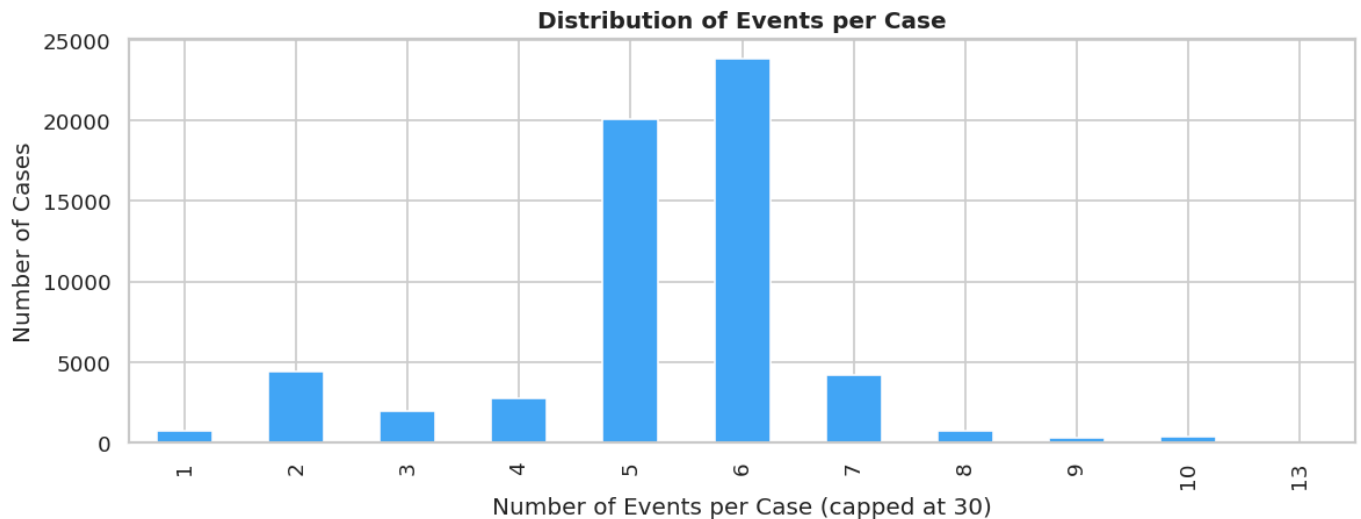


1.1 Activity Frequency

The 5 BPMN milestone activities (dark blue) dominate the log. Create Purchase Order Item appears in every case. Remove Payment Block (15,456) and Receive Order Confirmation (17,445) are the most common non-milestone activities systemic deviations from the ideal flow.



1.2 Events per Case & Case Complexity

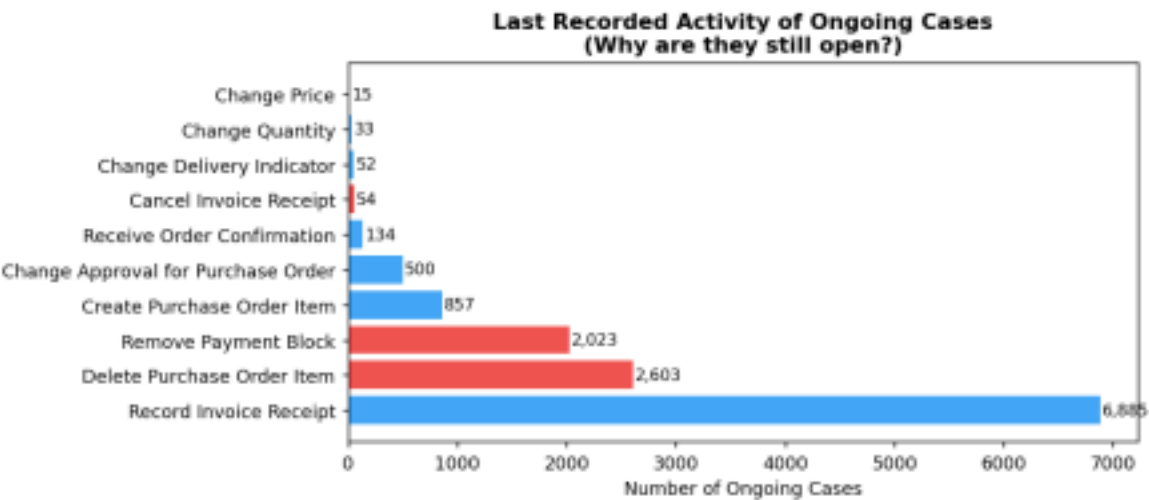


Most cases have 5 or 6 events. Cases with exactly 5 events follow the ideal flow perfectly. The dominant peak at 6 events (~23,800 cases) indicates one additional activity most commonly Receive Order Confirmation is extremely widespread.

- 1–4 events: ~8,500 cases Consignment or ongoing cases missing invoice activities
- 5 events: ~20,100 cases ideal flow candidates
- 6 events: ~23,800 cases one extra activity (mostly Receive Order Confirmation)
- 7+ events: ~5,000 cases complex cases with multiple extra activities

1.3 Completed vs Ongoing Cases Why Are Cases Still Open?

Per the project definition, 77.8% of cases are completed and 22.2% are ongoing. To understand why cases remain open, we inspected the last recorded activity of all 13,156 ongoing cases:



Last recorded activity of ongoing cases. Blue = normal mid-process; Red = problematic.

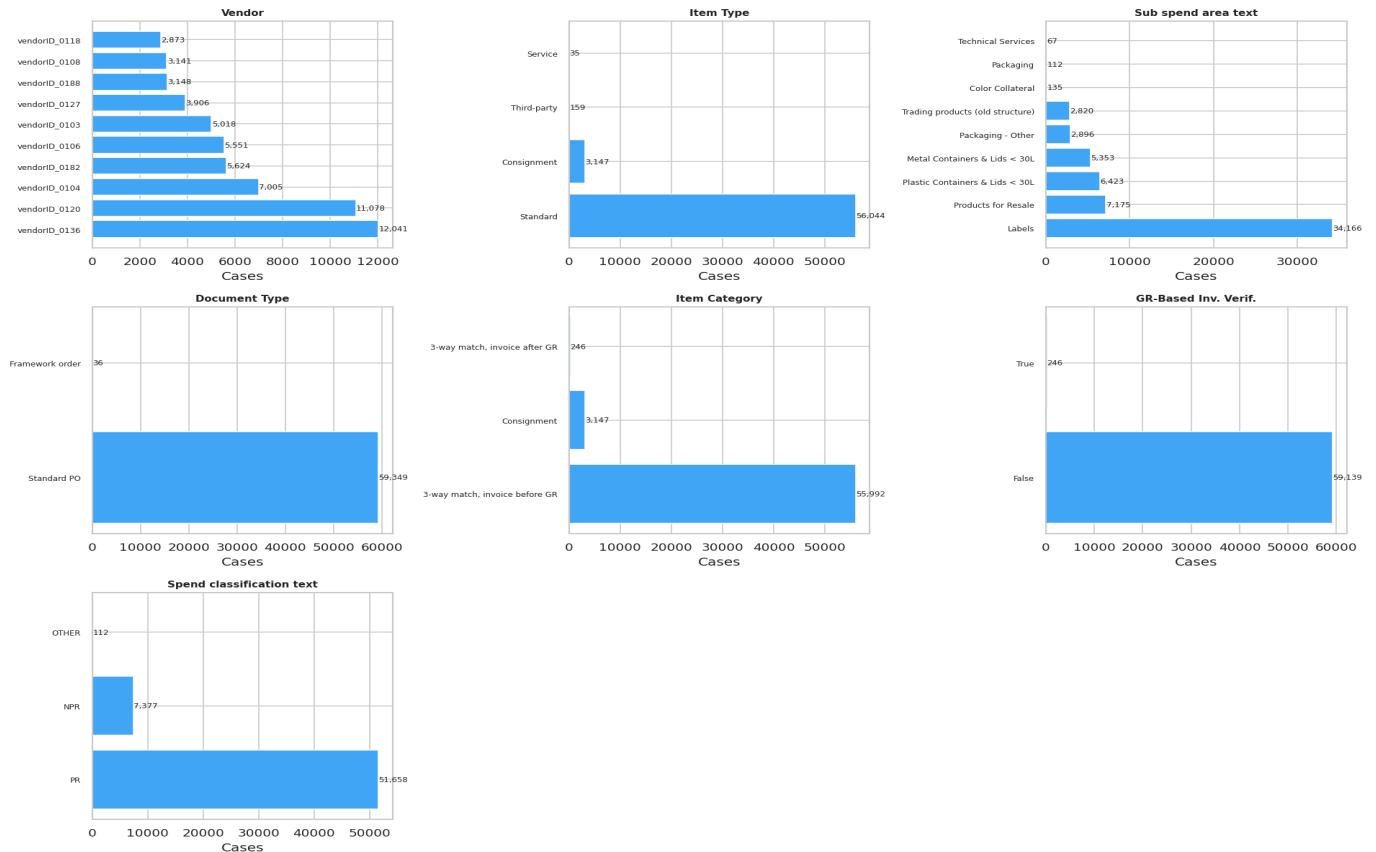
Last Activity	Ongoing Cases	Interpretation
Record Invoice Receipt	6,885	Waiting for invoice clearance stuck in the main bottleneck phase
Delete Purchase Order Item	2,603	Order deleted process terminated without clearing
Remove Payment Block	2,023	Payment blocked waiting for manual release
Create Purchase Order Item	857	Just started early stage, not yet problematic
Change Approval for Purchase Order	500	Approval pending 46% of these typically end in deletion

Other activities	288	Various mid-process states
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The largest group (6,885 cases = 52.3% of ongoing) is stuck at Record Invoice Receipt consistent with the 2.31-month bottleneck identified in Task 2. The 2,603 Delete PO Item cases represent terminated orders, not genuinely ongoing processes.

1.4 Categorical Attribute Distributions

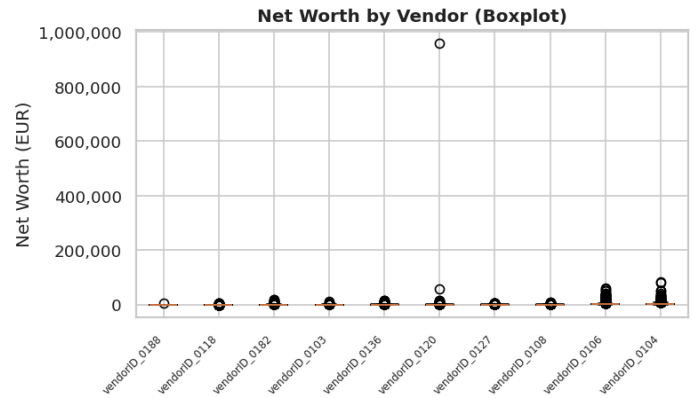
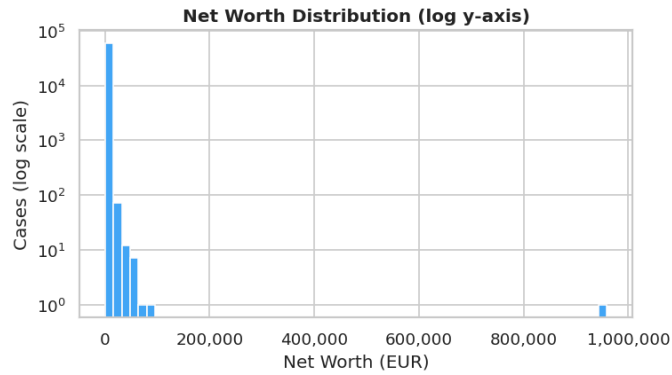
Case-Level Categorical Attribute Distributions



Case-level categorical distributions across 7 key attributes.

- **Vendor:** vendorID_0136 (12,041) and vendorID_0120 (11,078) handle the most cases. vendorID_0118 (2,873) handles fewest.
- **Item Type:** Standard dominates (94.4%, 56,044 cases). Consignment (5.3%, 3,147) follows a completely separate process.
- **Sub Spend Area:** Labels is dominant with 34,166 cases (57.5%). The top 4 spend areas cover 89% of all cases.
- **Item Category:** 55,992 cases (94.2%) use 3-way match invoice before GR. Only 246 use invoice after GR.
- **Document Type:** 59,349 Standard PO vs 36 Framework orders near constant, no analytical value.

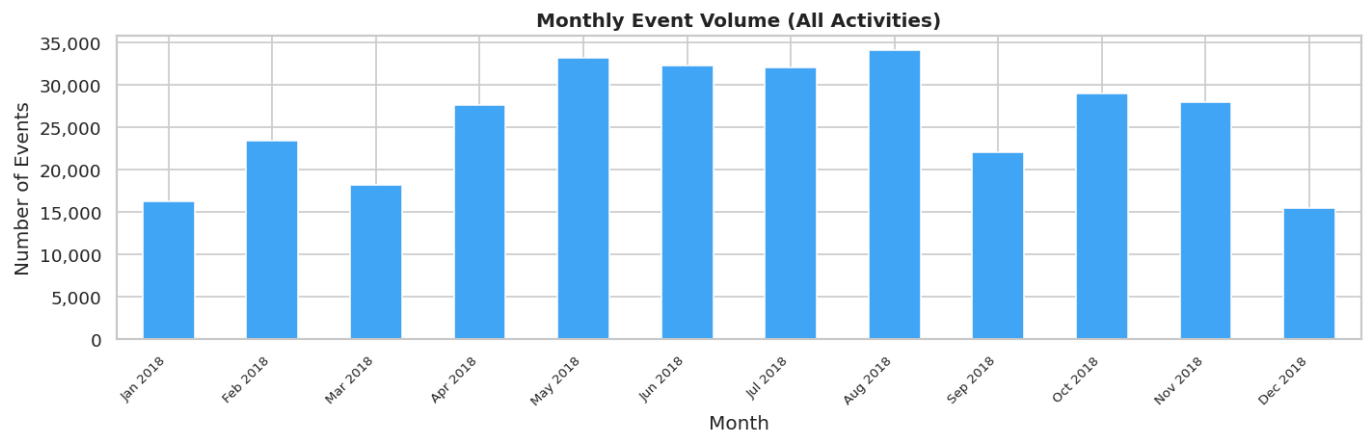
1.5 Net Worth Distribution



Net worth distribution (log scale) and by vendor.

Net worth is highly right-skewed. The vast majority of line items are below €50,000 with rare outliers reaching €1,000,000. vendorID_0120 has the highest single outlier (~€960,000). This skew means mean-based financial analysis would be misleading median is more appropriate for this dataset.

1.6 Temporal Coverage & Monthly Volume

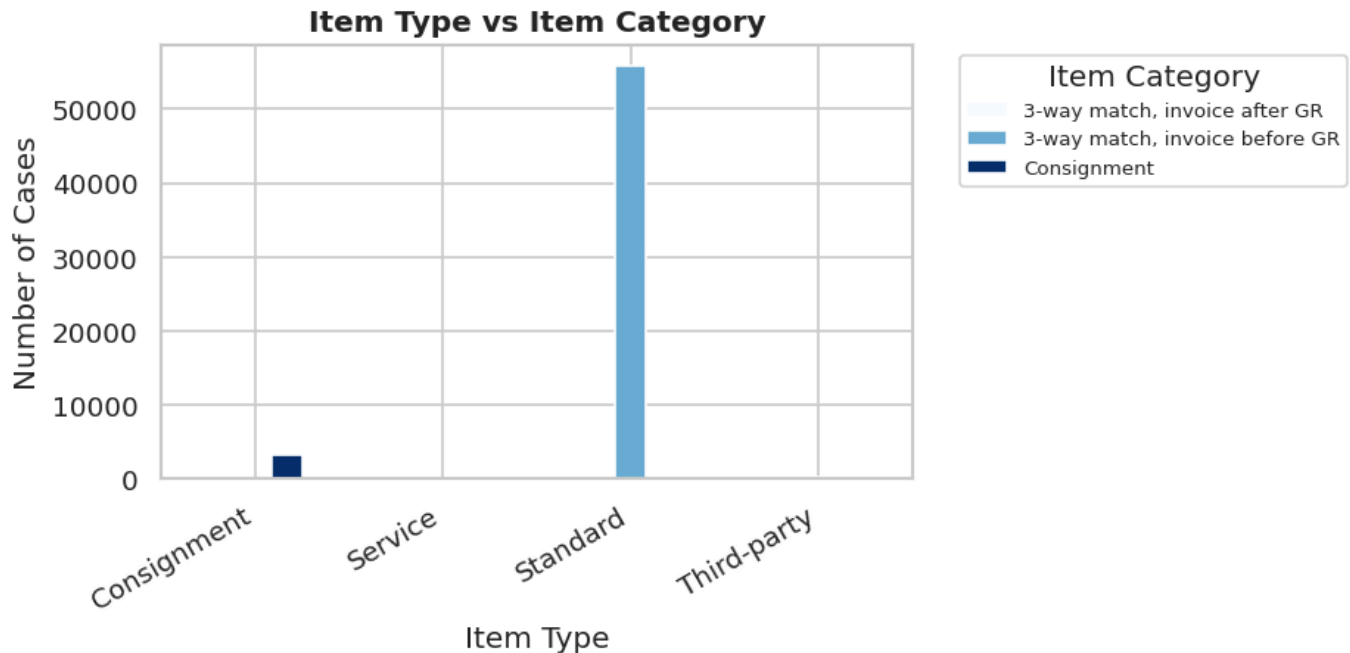


Monthly event volume. Clear seasonal pattern with August peak.

The log covers January to December 2018. Monthly event volume is not uniform: January (16,400), March (18,200), and December (15,600) are the quietest months, while August (34,000) and May (33,300) are peak months. This seasonal pattern is explored further in Task 6.

1.7 Column Relationships

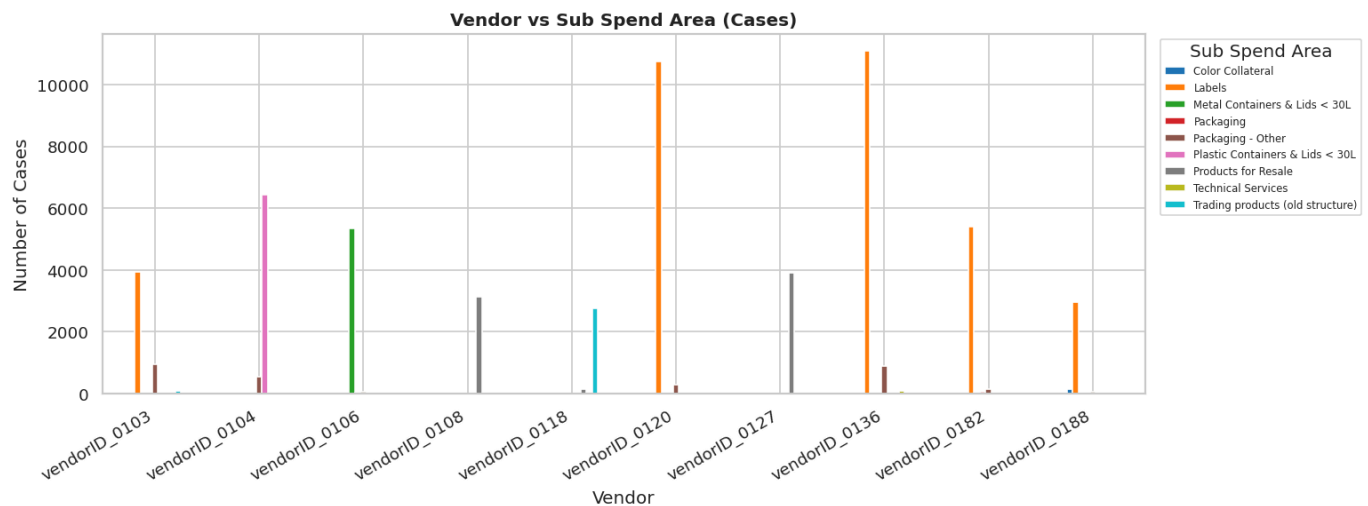
Relationship 1: Item Type ↔ Item Category



Item Type vs Item Category: near-perfect one-to-one mapping.

There is a near-perfect one-to-one mapping between Item Type and Item Category. Consignment items exclusively use the Consignment category. Standard items almost exclusively use "3-way match, invoice before GR" (55,992 cases) with only 246 using "invoice after GR". This means **Item Category** and **GR-Based Inv. Verif.** are both redundant with **Item Type** they carry no additional analytical information beyond what Item Type already tells us.

Relationship 2: Vendor ↔ Sub Spend Area



Vendor vs Sub Spend Area: strong specialisation pattern.

Each vendor exclusively handles one or two specific spend areas. This strong specialisation is critical context: vendorID_0120 and vendorID_0136 are the primary Labels vendors (10,700+ and 11,000+ cases). vendorID_0104 handles only Plastic Containers, vendorID_0106 handles only Metal Containers, vendorID_0108 and vendorID_0127 handle Products for Resale. This means vendor performance differences (Task 10) are confounded with spend area differences any vendor comparison must control for spend area.

Relationship 3: GR-Based Inv. Verif. ↔ Item Category

GR-Based Inv. Verif.	Item Category	Cases	Interpretation
False (59,139)	3-way match, invoice before GR	55,992	Invoice received before goods standard flow
False (59,139)	Consignment	3,147	No GR verification needed for consignment
True (246)	3-way match, invoice after GR	246	GR must occur before invoice can be verified

The GR-Based Inv. Verif. flag is True for exactly the 246 cases with "invoice after GR" category. These two columns are therefore completely redundant with each other only one needs to be retained.

1.8 Column Analysis Summary

Column	Role	Justification
Case Identifier	Key	Unique ID per line item. Never null, 59,385 unique values = one per case.
label	Key	21 unique activities. Distribution shows 5 milestones + 16 extra.
timestamp	Key	All events are lifecycle=complete. Range: Jan–Dec 2018.
Vendor	Key	10 unique IDs. Strongly mapped to spend area (see Figure 8).
Sub spend area text	Key	10 sub-categories. Labels = 57.5%. Used for SLA thresholds (Task 3).
Item Type	Key	Standard (94.4%), Consignment (5.3%), Third-party (0.3%), Service (<0.1%).

Cumulative net worth (EUR)	Key	Highly right-skewed. Range: €0 – €960,000. Median more useful than mean.
Purchasing Document	Supporting	Multiple line items share one PO. 59,385 cases from ~9,118 documents.
Spend area text	Supporting	Parent category of Sub spend area. 3 values: Packaging, Trading, Sales.
Document Type	Supporting	Standard PO (99.9%) vs Framework order (36 cases). Near-constant.
Spend classification text	Supporting	PR (86.7%), NPR (12.4%), OTHER (0.2%).
GR-Based Inv. Verif.	Redundant	True for exactly the 246 "invoice after GR" cases. Fully redundant with Item Category.
Source	Redundant	Always "sourceSystemID_0000". Single ERP system zero analytical value.
Company	Redundant	Always "companyID_0000". Single company zero analytical value.
Goods Receipt	Redundant	Always "true" no variation whatsoever.
Purch. Doc. Category name	Redundant	Always "Purchase order" zero variation.

lifecycle:transition	Redundant	Always "complete" no start events recorded.
User / resource	Redundant	Both columns are identical (verified programmatically).
Name	Redundant	Duplicate of Vendor column same values.

Key Findings

- **Redundant columns:** Source, Company, Goods Receipt, lifecycle: transition, User=resource, Name=Vendor all constant values, safely removable.
- **Consignment (5.3%):** No invoice activities completely separate process. Must be excluded from all invoice-related tasks.
- **Labels dominates:** 34,166 cases (57.5%), the single most important spend category.
- **Net worth is right-skewed:** Most items below €50K with rare outliers up to €1M.
- **Seasonal pattern:** August peak (34,000 events), December dip (15,600 events).
- **Vendor specialisation:** Each vendor handles 1–2 specific spend areas exclusively.

Column Summary

Column	Role	Interpretation
Case Identifier	Key	Unique ID per line item (PurchDoc + Item)
label	Key	21 activities 5 milestones + 16 change/cancel/admin
timestamp	Key	Datetime of activity completion
Vendor	Key	10 anonymized IDs main dimension for Task 10
Sub spend area text	Key	10 sub-categories used for SLA thresholds (Task 3)
Item Type	Key	Standard (94.4%), Consignment (5.3%), Third-party, Service
Source / Company / Goods Receipt	Redundant	All constant values zero analytical utility
User / resource	Redundant	Identical duplicate columns

Improvement Suggestions

- Remove 5 redundant columns from all future analysis constant values add noise.
- Treat Consignment as a separate process filter from all invoice-related tasks.
- Investigate 22.2% ongoing cases some may be stalled beyond expected cycle times.
- Investigate Remove Payment Block (15,456 cases) systemic bottleneck identified in Task 2.

Task 2: Understand the Process

Procedure

Process understanding used Apromore Process Discoverer (visual maps, frequency and performance overlays) and Python (variant statistics, cycle time analysis). Full event log: 311,821 events, 59,385 cases, 21 activities, Jan–Dec 2018.

Key Figures

2.46 mths Avg Duration Jan–Dec 2018	Instant 11.47 mths Range full spread	221 Variants unique sequences	~39% Ideal Flow completed non-consignment
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Process Map All Paths (Full View)

Figure 2 shows the full process map with all 41 nodes and 99 arcs visible. It reveals the full complexity of the actual process 221 unique variants, multiple back-loops, and many paths bypassing the sequential milestone order. Dashed arrows = rare paths; thick solid arrows = dominant flow.

- 221 unique case variants across 59,385 cases process is far more complex than the ideal 5-activity model
- Multiple back-loops visible Record Goods Receipt and Vendor creates invoice can be revisited (rework)
- Many paths bypass sequential order Clear Invoice reachable directly from multiple earlier activities
- Dashed arrows = rare/infrequent paths; thick solid arrows = dominant high-frequency flow

The filtered map (Nodes=20, Arcs=20) reveals the dominant process backbone. The most common deviations: Receive Order Confirmation (17,445 cases, 29.4%) and Remove Payment Block (15,456 cases, 26.0%) neither in the ideal BPMN model.

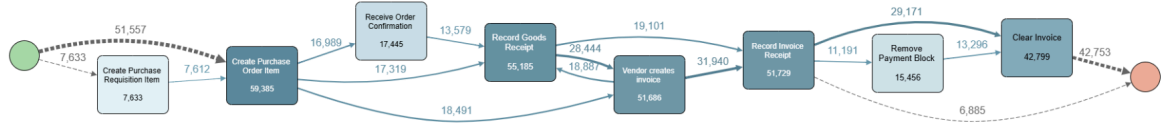


Figure 3 Filtered process map. Dominant backbone clearly visible.

Activity	Cases	Note
Create Purchase Order Item	59,385	Mandatory first milestone
Receive Order Confirmation	17,445	Non-milestone very common (29.4%)
Record Goods Receipt	55,185	Milestone parallel with Vendor invoice
Vendor creates invoice	51,686	Milestone parallel with Goods Receipt
Record Invoice Receipt	51,729	Milestone after both GR and invoice
Remove Payment Block	15,456	Non-milestone common extra step (26.0%)
Clear Invoice	42,799	Final milestone

Bottleneck Analysis Duration View

All activity nodes execute instantly delays are entirely in waiting time between activities. The invoice clearance phase is the dominant bottleneck.

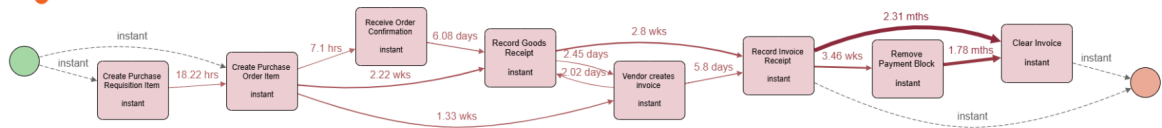
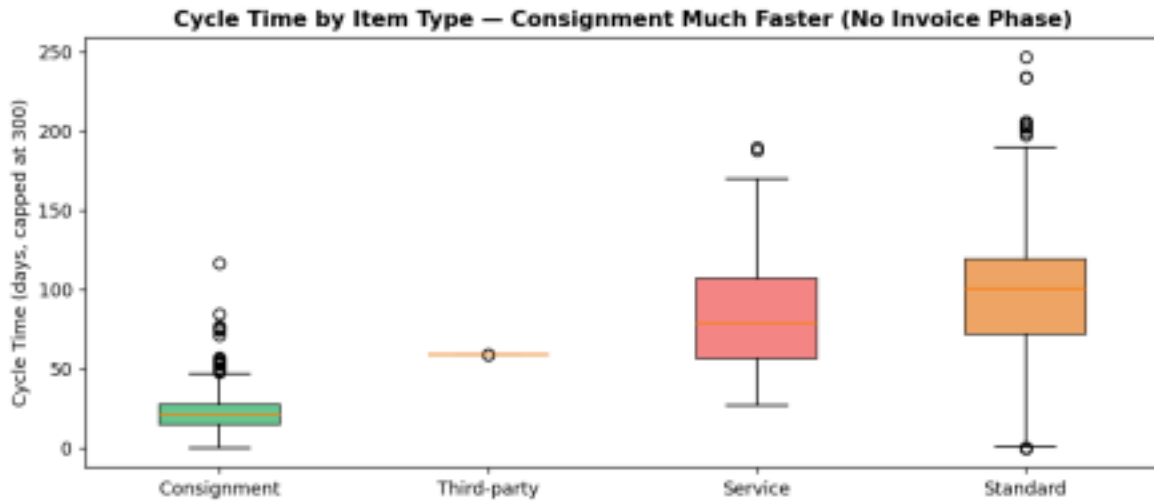


Figure 4 Duration overlay. Darker red = longer wait = bottleneck.

From	To	Avg Duration	Severity
Record Invoice Receipt	Clear Invoice	2.31 months	Critical
Remove Payment Block	Clear Invoice	1.78 months	Critical

Record Invoice Receipt	Remove Payment Block	3.46 weeks	Critical
Record Goods Receipt	Record Invoice Receipt	2.8 weeks	High
Vendor creates invoice	Record Invoice Receipt	5.8 days	Moderate

2.4 Process Variant Comparison by Item Type

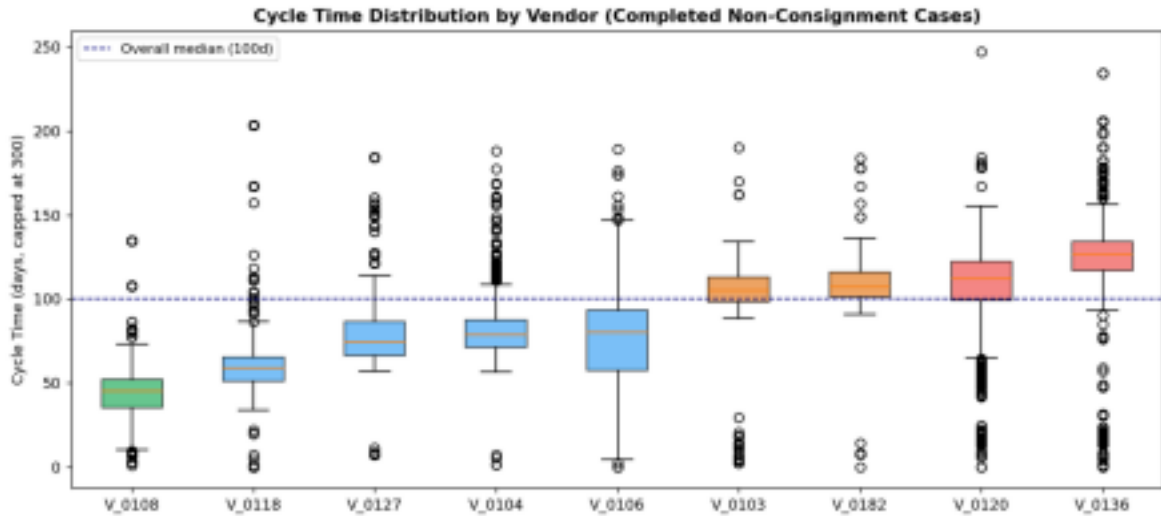


Cycle time by item type. Consignment is structurally different and much faster.

Item Type	Cases	Median CT (days)	Mean CT (days)	Key Observation
Consignment	2,848	21.2	22.4	No invoice phase process ends at goods receipt. Very fast.
Third-party	159	59.1	59.1	Few cases. Faster than Standard simpler process.
Service	27	79.0	90.1	Very few cases. Similar to Standard.
Standard	43,195	100.9	95.4	Dominant type. Full 5-milestone process including invoice clearance.

The 79-day gap between Consignment (21.2 days) and Standard (100.9 days) directly confirms that the invoice clearance phase (P3) present only in Standard items is the dominant driver of cycle time. Consignment cases reach their natural end at Record Goods Receipt with no invoice activities.

2.5 Process Variant Comparison by Vendor



Cycle time by vendor (completed non-consignment). Navy dashed = overall median (99 days).

Vendor	Median CT (days)	Mean CT (days)	Cases	Assessment
vendorID_0108	45.2	44.1	2,856	Best performer consistently fast
vendorID_0118	59.1	64.0	2,795	Good well below overall median
vendorID_0127	74.2	82.9	2,877	Above average performance
vendorID_0104	78.7	80.7	5,532	Near median acceptable
vendorID_0106	80.3	77.7	4,231	Near median acceptable
vendorID_0103	105.1	101.7	3,198	Below average needs attention
vendorID_0182	107.2	107.8	4,091	Below average needs attention
vendorID_0120	112.2	106.6	8,771	Slow high volume makes this impactful
vendorID_0136	126.0	123.3	9,030	Worst performer high volume + slow

There is a 3x difference in median cycle time between the fastest (vendorID_0108, 45.2 days) and slowest (vendorID_0136, 126.0 days) vendor. Since each vendor specialises in specific spend areas (established in Task 1), the differences partly reflect spend area complexity but also genuine vendor-specific performance variation explored further in Task 10.

2.6 Improvement Suggestions

- **Address payment blocks (highest priority).** Remove Payment Block affects 26% of cases and adds up to 1.78 months delay. Implementing tolerance-based 3-way matching (PO vs GR vs invoice within $\pm 2\%$) would automate block release and eliminate the majority of this delay.
- **Standardise order confirmation.** Receive Order Confirmation in 17,445 cases adds overhead that is not in the ideal model. Formalising it as an optional milestone with an SLA target would at minimum make it measurable.
- **Root**

cause: invoice clearance. The 2.31-month average from Record Invoice Receipt to Clear Invoice is the single largest delay. This likely reflects finance team capacity constraints or approval processes dedicated investigation and SLA setting is recommended.

- **Vendor performance review.** vendorID_0136 (9,030 cases, median 126 days) and vendorID_0120 (8,771 cases, median 112 days) are the two slowest high-volume vendors. Given their case volume, improving these two vendors alone would significantly reduce overall average cycle time.

Task 3 SLA Compliance

Procedure

SLA compliance was evaluated using Python on the shared case-level dataset from Task 1. Only completed non-Consignment cases included (43,381). Ongoing cases excluded incomplete cycle times would underestimate violations.

SLA Thresholds & Overall Compliance

Spend Area	SLA
Labels	15 weeks (105 days)
Products for Resale	13 weeks (91 days)
Plastic & Metal Containers & Lids < 30L	16 weeks (112 days)
All others	11 weeks (77 days)

43,381 Cases Analysed completed non-consignment	23,123 SLA Met 53.3%	20,258 SLA Violated 46.7% critical
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With 46.7% of cases exceeding their SLA threshold, CoatXR's SLAs are not being met. This is a critical performance issue requiring urgent attention.

Violations per Spend Area

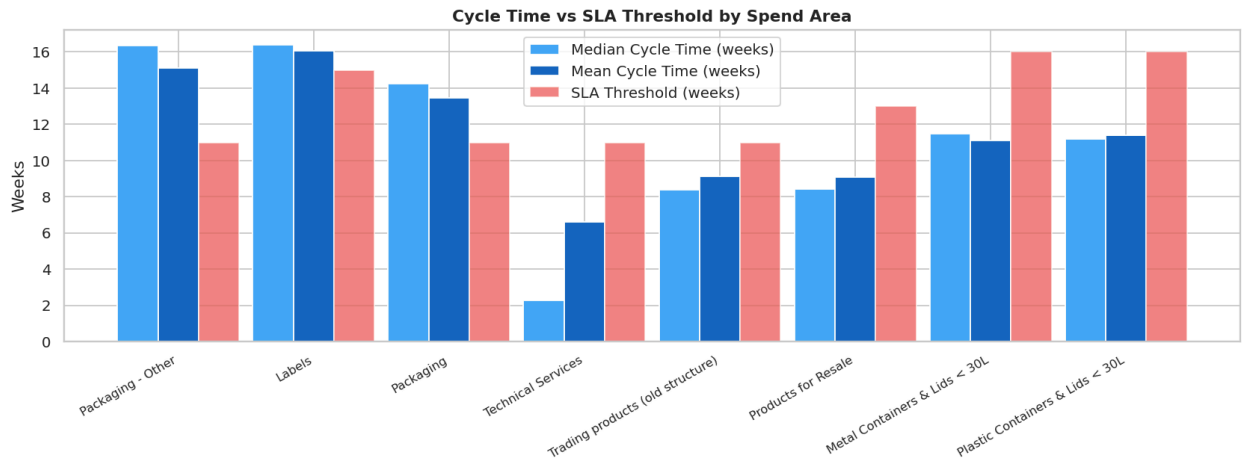


Figure 4 : SLA violation rate (left) and absolute violations (right) per spend area.

Spend Area	SLA (wks)	Cases	Violations	Rate	Median (wks)
Packaging - Other	11	882	789	89.5%	16.3
Labels	15	24,349	18,085	74.3%	16.4
Packaging	11	54	40	74.1%	14.2

Technical Services	11	19	6	31.6%	2.3
Trading products	11	2,686	294	10.9%	8.4
Products for Resale	13	5,855	639	10.9%	8.4
Metal Containers < 30L	16	4,180	303	7.2%	11.5
Plastic Containers < 30L	16	5,299	68	1.3%	11.2

Cycle Time Distributions

... Top 10 worst SLA violations (by days over threshold):

	Case Identifier	Sub spend area text	Vendor	cycle_weeks	sla_weeks	days_over	weeks_over
16290	4507021966_00120	Labels	vendorID_0120	35.30	15.00	141.98	20.30
19142	4507024785_00170	Labels	vendorID_0136	33.40	15.00	129.10	18.40
19158	4507024785_00340	Labels	vendorID_0136	33.40	15.00	129.10	18.40
19165	4507024785_00410	Labels	vendorID_0136	33.40	15.00	129.10	18.40
15519	4507021136_00020	Trading products (old structure)	vendorID_0118	29.00	11.00	126.08	18.00
15520	4507021136_00040	Trading products (old structure)	vendorID_0118	29.00	11.00	126.08	18.00
15521	4507021136_00050	Trading products (old structure)	vendorID_0118	29.00	11.00	126.08	18.00
15522	4507021136_00060	Trading products (old structure)	vendorID_0118	29.00	11.00	126.08	18.00
15523	4507021136_00070	Trading products (old structure)	vendorID_0118	29.00	11.00	126.08	18.00
15524	4507021136_00080	Trading products (old structure)	vendorID_0118	29.00	11.00	126.08	18.00

Figure 5 Cycle time per spend area. Red dashed line = SLA threshold.

- **Labels and Packaging - Other:** Entire interquartile range above SLA systematic failure, not outliers.
Labels = 89.3% of ALL violations.
- **Containers categories:** Boxes comfortably below SLA line healthy performance.
- **Technical Services:** Wide spread outlier problem, not systemic.

Improvement Suggestions

- **Packaging - Other (89.5%):** Investigate root cause nearly all cases fail. Redesign process or revise SLA.
- **Labels (74.3%):** Highest absolute impact. Reducing invoice clearance delays (Task 2 bottleneck) would directly improve SLA compliance here.
- **Vendor review:** vendorID_0136 and vendorID_0118 appear in worst violations targeted performance reviews recommended.
- **Focus on Labels.** 57.5% of all cases, 89.3% of all violations improving Labels alone transforms overall SLA compliance.

Task 4 Deviations from Ideal Flow

Procedure

The ideal process consists of exactly five milestone activities: Create Purchase Order Item, Record Goods Receipt, Vendor creates invoice (the latter two in any order), Record Invoice Receipt, and Clear Invoice. Consignment cases were excluded as they follow a separate unmodeled process. For each remaining case we checked whether its activity sequence consisted solely of these five milestones in the prescribed order. Cases were split into completed (last event = Clear Invoice or Record Goods Receipt) and ongoing, since missing milestones in ongoing cases reflect incompleteness, not deviation.

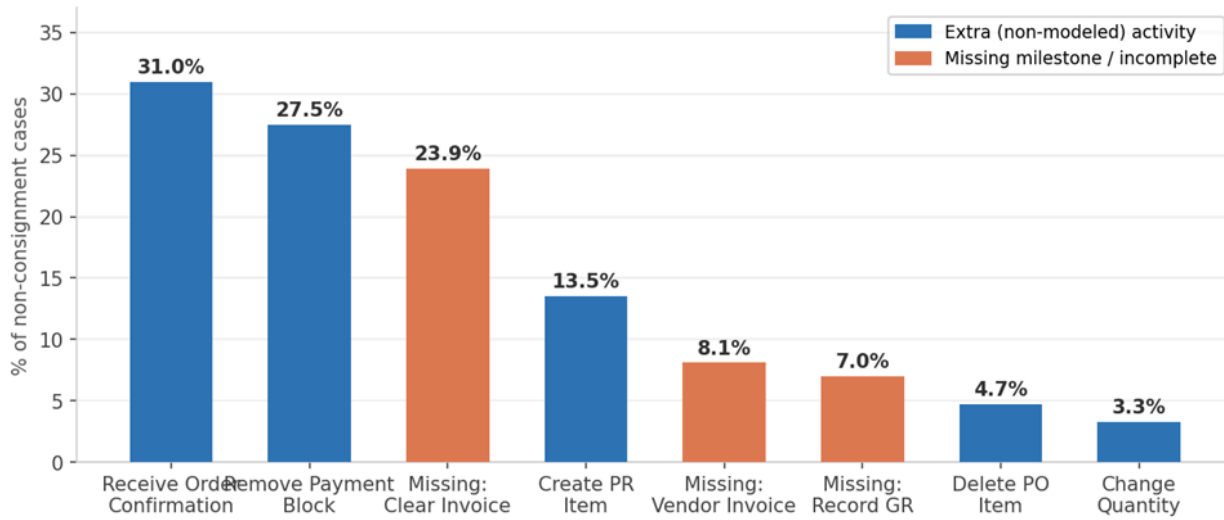
Key Figures

Non-consignment cases	Follow ideal flow (all)	Follow ideal flow (completed)	Completed cases
56,238	30.1%	39.0%	43,381
total scope	16,905 cases	of 43,381 completed	77.2% of scope

Deviation Frequency

The chart below shows the eight most common deviation types as a percentage of all non-consignment cases. Blue bars represent extra (non-modeled) activities; coral bars represent missing milestones.

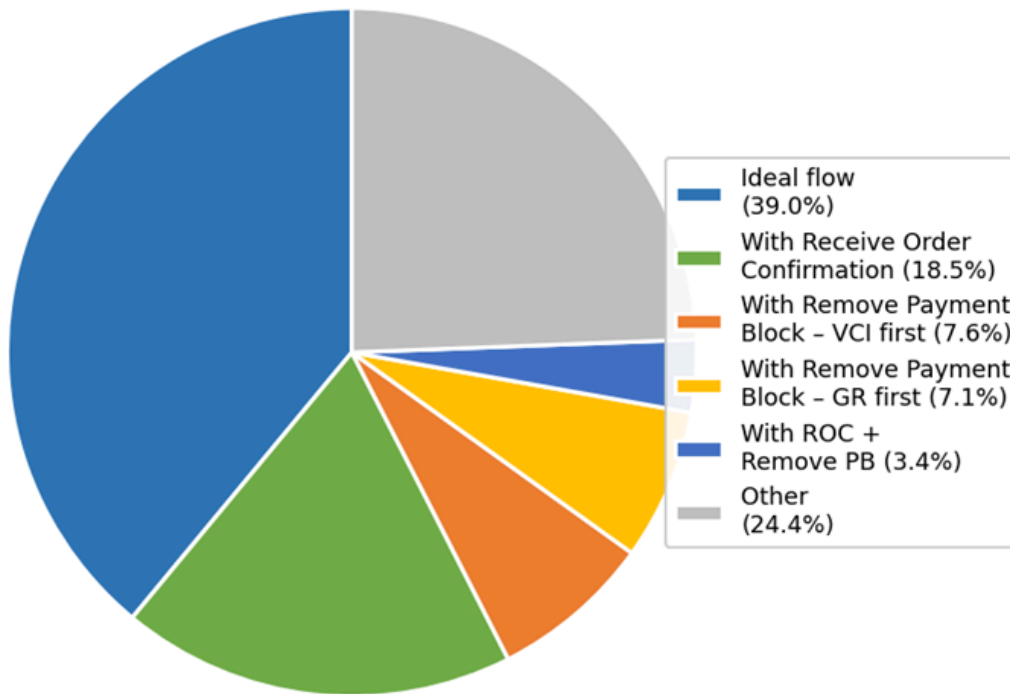
Figure 1 — Frequency of each deviation type across non-consignment cases



Process Variant Distribution

Among completed non-consignment cases, the ideal flow (both valid milestone orderings combined) accounts for 39% of cases.

Figure 2 — Distribution of top process variants
(completed non-consignment cases)



Findings

- **Receive Order Confirmation (31%)** legitimate vendor acknowledgement not in the ideal model; not an error but adds complexity.
- **Remove Payment Block (27.5%)** most operationally concerning deviation; signals invoice discrepancies requiring manual clearance before settlement.
- **Missing Clear Invoice (23.9%)** largely explained by ongoing cases, not a structural problem.
- **Create Purchase Requisition Item (13.5%)** cases originating from a purchase requisition, adding a pre-PO step.
- **Delete Purchase Order Item (4.7%)** full order terminations, analysed in Task 8.
- **Change Quantity (3.3%)** late order corrections, analysed in Task 9.

Improvement Suggestions

Automate invoice matching to reduce the 27.5% of cases requiring a manual Remove Payment Block. Implementing tolerance-based 3-way matching (PO vs. GR vs. invoice) would eliminate most payment blocks without human intervention.

Formalise Receive Order Confirmation in the process model. Since it occurs in 31% of cases, treating it as an optional but modelled milestone would improve transparency and enable SLA monitoring for vendor response times.

Introduction

Tasks 5-7 for CoatXR P2P (59,385 cases, 311,821 events, 2018-01-01 to 2018-12-31).

Completed: 46,229;

Ongoing: 13,156.

Task 5 - Process milestones and phases

Procedure

The analysis was performed on completed cases only (46,229 cases). Ongoing cases were excluded because they have not yet reached their final milestone, meaning phase durations would be incomplete and would underestimate the true time spent in each phase.

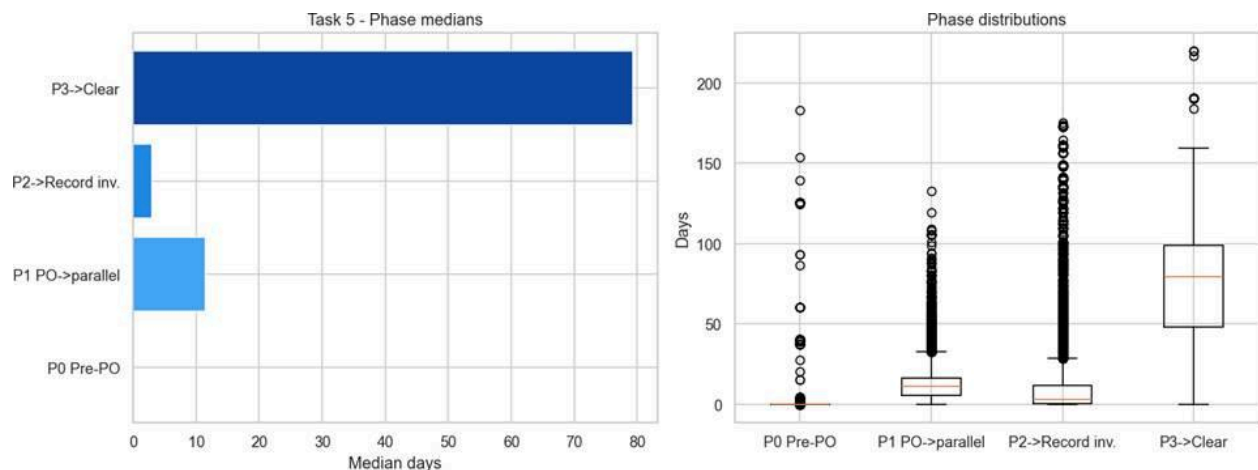
The five milestone activities are:

1. Create Purchase Order Item
2. Record Goods Receipt
3. Vendor creates invoice
4. Record Invoice Receipt
5. Clear Invoice
6. **P0 (Pre-PO)**: From case start to the first occurrence of Create Purchase Order Item.

This phase captures any pre-processing before the purchase order is formally created, such as requisition steps.

7. **P1 (PO to Parallel End):** From the last occurrence of Create Purchase Order Item to the later of the last Record Goods Receipt and last Vendor creates invoice. This captures the time to both receive the goods and obtain the vendor invoice whichever takes longer defines the end of this parallel phase.
8. **P2 (Parallel End to Invoice Receipt):** From the end of the parallel phase to the last occurrence of Record Invoice Receipt. This is the administrative step of formally recording the invoice in the system after both goods and invoice are available.
9. **P3 (Invoice Receipt to Clear Invoice):** From the last Record Invoice Receipt to the last Clear Invoice. This is the payment clearance phase the final step before the case closes.

We use the **last occurrence** of each milestone as the phase boundary. This is because some milestones are repeated due to rework (e.g. Record Goods Receipt can occur more than once if a partial delivery happens). Using the last occurrence ensures the phase covers the full duration of that activity including any rework, rather than measuring only up to the first attempt.



Finding: P3 clearing dominates. Recommend: reduce payment/clearing delays and invoice cancellations.

Key Figures

Phase	Description	Median Duration
P0	Pre-PO (case start → Create PO Item)	0.0 days
P1	PO → Parallel end (GR + Vendor Invoice)	11.41 days
P2	Parallel end → Record Invoice Receipt	2.97 days
P3	Record Invoice Receipt → Clear Invoice	79.28 days

Findings

P0 (0.0 days median): The vast majority of cases begin directly with Create Purchase Order Item there is effectively no pre-PO phase in the typical case. Cases that do have a pre-PO step (such as Create Purchase Requisition Item, which occurs in 7,633 cases) add a small delay at the start, but this is negligible at the median level.

P1 (11.41 days median): This phase covers the parallel execution of goods delivery and vendor invoicing. 11 days is a reasonable lead time for receiving goods and obtaining the vendor invoice simultaneously. The spread is wide (as shown in the boxplot) indicating high variability some vendors are much slower than others at delivering goods or submitting invoices.

P2 (2.97 days median): The administrative recording of the invoice receipt is fast typically under 3 days. This is a largely automated or simple clerical step with low variance.

P3 (79.28 days median main bottleneck): This phase completely dominates the process. At a median of 79.28 days, P3 accounts for approximately 87% of the total median cycle time (90.8 days for completed cases). This means once the invoice receipt is recorded, it takes over 2.5 months on average before the invoice is actually cleared for payment.

This finding is fully consistent with the bottleneck identified in Task 2, where the transition from Record Invoice Receipt to Clear Invoice showed an average duration of 2.31 months in the Apromore duration overlay.

Activities within P3: The P3 phase contains several non-milestone activities that contribute to its length, most notably Remove Payment Block (15,456 occurrences). When a payment block is applied, the invoice cannot be cleared until it is manually released this alone accounts for a significant portion of the 79-day P3 median.

Improvement Suggestions

- **Reduce payment blocks in P3.** Since Remove Payment Block is the primary cause of delay in P3, automating block release for cases where the invoice matches the PO within a tolerance (e.g. $\pm 2\%$) would significantly reduce the 79-day median.
- **Monitor P1 by vendor.** The wide spread in P1 suggests vendor-specific delivery and invoicing speed differences. Vendors with consistently slow P1 durations should be prioritised for SLA negotiation.
- **Investigate invoice cancellations in P3.** Cases with Cancel Invoice Receipt (571 cases) add significant delay within P3, as they require re-processing of the entire invoice step.

Task 6 - WiP and seasonality

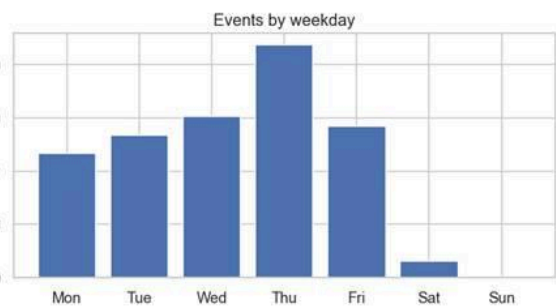
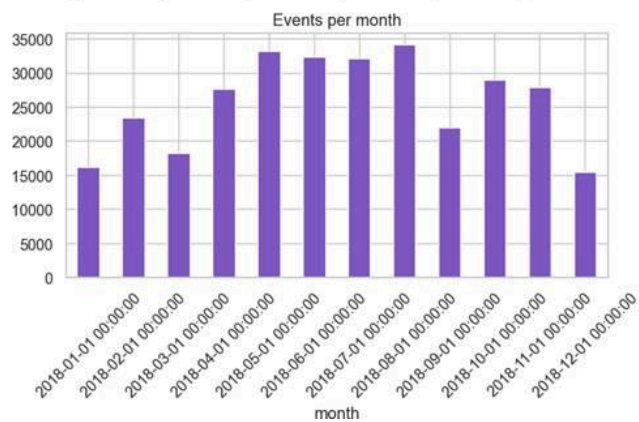
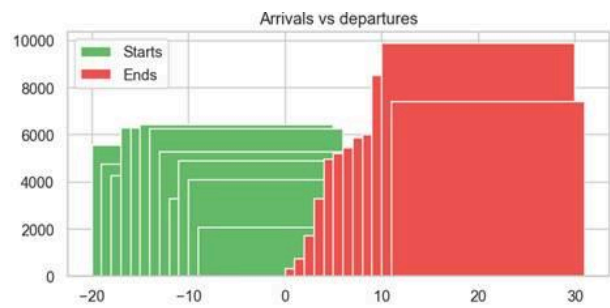
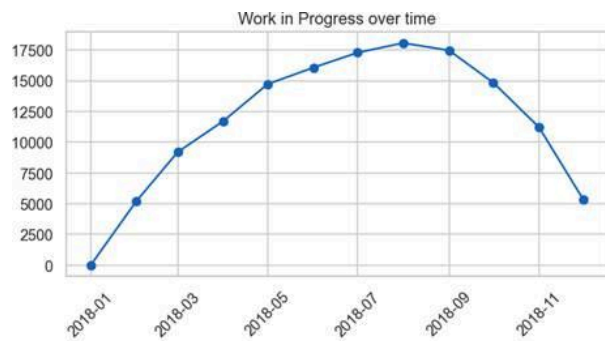
Procedure

Work in Progress (WiP) was calculated for each month of 2018 by counting the number of cases that had started but not yet completed at the beginning of each month. Arrivals were counted as cases whose start month matched the target month. Departures were counted as cases whose end month matched the target month. The monthly event volume and weekday event distribution were also analysed to identify seasonality patterns.

Key Figures WiP Over Time

Month	WiP	Arrivals	Departures	Net Change
Jan 2018	0	5,546	341	+5,205
Feb 2018	5,205	4,751	756	+3,995
Mar 2018	9,200	4,250	1,718	+2,532
Apr 2018	11,732	6,299	3,299	+3,000

May 2018	14,732	6,305	4,959	+1,346
Jun 2018	16,078	6,412	5,195	+1,217
Jul 2018	17,295	6,243	5,461	+782
Aug 2018	18,077	5,263	5,864	-601
Sep 2018	17,476	3,279	6,003	-2,724
Oct 2018	14,849	4,894	8,510	-3,616
Nov 2018	11,223	4,083	9,869	-5,786
Dec 2018	5,350	2,060	7,410	-5,350



Findings

WiP is not constant. It follows a clear arc across the year rising from zero in January to a peak of 18,077 in August, then declining sharply through to December.

Phase 1 Buildup (January to August): In every month from January through July, arrivals exceed departures (positive net change). New cases are being created faster than existing cases are being completed. January is particularly extreme 5,546 new cases arrive but only 341 complete, a net increase of +5,205. This reflects the start of the procurement year when new orders are placed but the process has not yet had time to complete them. WiP grows steadily as the backlog accumulates.

Phase 2 Peak (August): WiP reaches its maximum of 18,077 in August. At this point arrivals (5,263) and departures (5,864) are approximately balanced with departures slightly exceeding arrivals this is the turning point where the system begins to catch up with its backlog.

Phase 3 Catch-up (September to December): From September onwards, departures consistently and increasingly exceed arrivals. The system is completing more cases than it is receiving. By December, departures (7,410) are 3.6x arrivals (2,060), rapidly clearing the remaining backlog.

Sources of seasonality:

- The January–March low arrival volume likely reflects the start of the fiscal year when new procurement budgets are being set and approved.
- The April–July peak in arrivals (6,243–6,412 per month) represents the main procurement season.
- The August–December decline in arrivals reflects the wind-down of the procurement year combined with holiday periods in late December.
- The high departure volume in October–December reflects the completion of cases that were started mid-year and have now worked through the full process cycle (median cycle time ~99 days means cases started in July–August complete in October–November).

Weekday seasonality: The events-by-weekday chart shows a clear business-hours pattern. Thursday is the busiest day, followed by Wednesday and Friday. Saturday has minimal activity and Sunday has virtually none. This confirms the process is entirely human-driven with no automated weekend processing. The Thursday peak may reflect weekly batch processing or approval cycles that accumulate through the week.

Improvement Suggestions

- **Plan capacity for the August WiP peak.** With 18,077 cases simultaneously active in August, this is when processing delays and payment block backlogs are most likely to

occur. Additional staffing or automated processing during peak months would reduce the SLA violations concentrated in this period.

- **Implement monthly WiP monitoring.** The arrivals vs departures balance provides an early warning signal. When net change turns positive for multiple consecutive months, management should investigate whether cycle times are increasing.
- **Investigate the January ramp-up.** The extreme imbalance in January (5,546 arrivals, 341 completions) suggests a procurement burst at year start. Smoothing order creation across the year would prevent the WiP accumulation that drives the August peak.

Task 7 - Rework loops

Procedure

A rework loop is defined as one or more activities that occur more than once in the same case. For each case, all activity sequences were scanned and any repeated activity (or sequence of activities returning to a previously visited activity) was identified as a loop. Cases with at least one rework loop were counted, and the top 3 most frequent loops were selected for detailed analysis. For each loop, the median cycle time of cases containing that loop was compared against the median cycle time of all cases without that loop to quantify the impact.

Key Figures

- **Cases with any rework:** 1,179 out of 59,385 total (2.0%)
- **Overall median cycle time (completed cases):** 90.8 days

Top 3 Rework Loops

Loop 1 Record Goods Receipt → Record Goods Receipt

- **Frequency:** 346 cases
- **Cycle time with loop:** ~79.7 days median
- **Cycle time without loop:** ~80.0 days median
- **Impact:** -0.3 days (negligible)

This loop occurs when goods are received in multiple separate deliveries for example a partial shipment followed by a second delivery. Since the loop does not require any invoice

reprocessing, the cycle time impact is negligible (± 0.3 days). This is considered a normal operational pattern for split deliveries and is not a process quality concern.

Loop 2 Record Invoice Receipt → Record Invoice Receipt

- **Frequency:** 336 cases
- **Cycle time with loop:** ~82.4 days median
- **Cycle time without loop:** ~80.0 days median
- **Impact:** +2.4 days

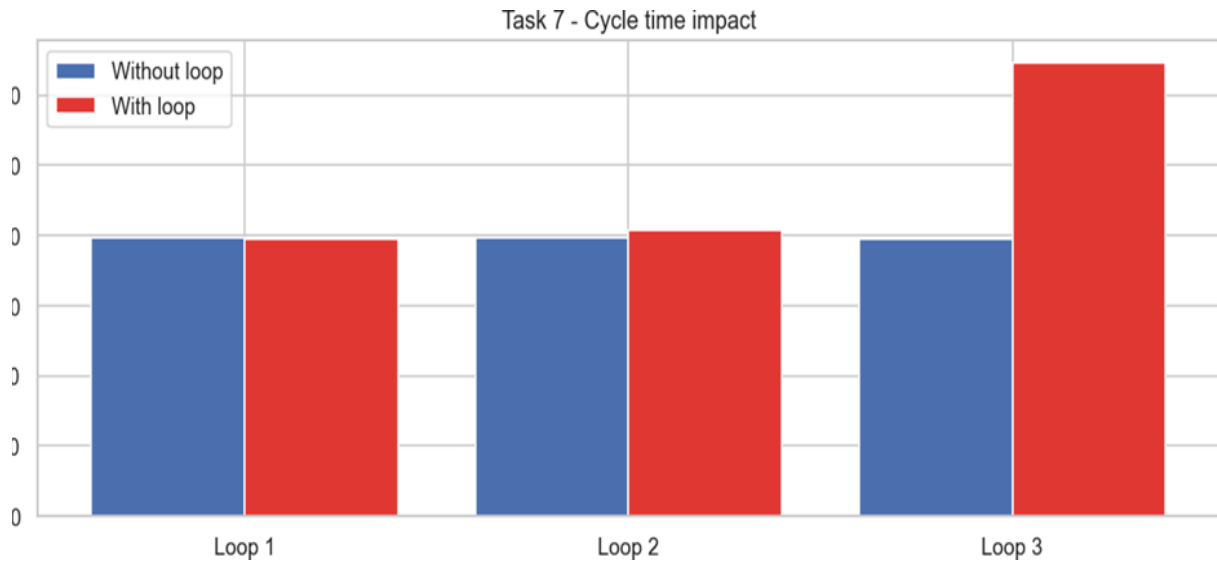
This loop occurs when an invoice receipt is recorded twice typically due to a minor correction or system entry error requiring re-entry. The 2.4-day impact is modest, suggesting the re-recording happens quickly. However, it still represents unnecessary manual rework that could be eliminated through better invoice entry validation.

Loop 3 Record Invoice Receipt → Cancel Invoice Receipt → Record Invoice Receipt

- **Frequency:** 245 cases
- **Cycle time with loop:** ~130.0 days median
- **Cycle time without loop:** ~80.0 days median
- **Impact:** +50.0 days**

This is by far the most severe rework loop. When an invoice receipt is cancelled and must be fully re-recorded, the case takes 50 days longer on average. The cancellation is triggered in 90.5% of cases by a vendor debit memo meaning the vendor submitted an incorrect invoice, requiring full reversal and reissue. After cancellation, the process must restart the entire invoice phase: a new Record Invoice Receipt is needed, often followed by Remove Payment Block before the case can finally reach Clear Invoice.

The 50-day penalty represents a major process inefficiency concentrated in a small number of cases (245 cases = 0.4% of all cases) but with severe individual impact.



Findings

Rework is rare overall only 2.0% of cases contain any repeated activity. However when rework does occur, the impact varies dramatically:

Loop	Cases	Frequency	Cycle Time Impact
Record GR → Record GR	346	0.58%	+0.0 days (negligible)
Record IR → Record IR	336	0.57%	+2.4 days (minor)
Record IR → Cancel IR → Record IR	245	0.41%	+50.0 days (severe)

Loop 1 is harmless split deliveries are a normal business pattern. Loop 2 is a minor quality issue. Loop 3 is the critical finding a 50-day penalty caused entirely by vendor billing errors that require invoice cancellation and reprocessing.

Improvement Suggestions

- Prevent invoice receipt cancellations (Loop 3 highest priority).** Since 90.5% of Cancel Invoice Receipt events are linked to vendor debit memos, the root cause is vendor-side invoice errors. Implementing vendor portal auto-validation where the vendor's invoice is automatically checked against the PO and GR before submission would prevent incorrect invoices from entering the system in the first place, eliminating most Loop 3 occurrences and recovering 50 days per affected case.

- **Add invoice entry validation for Loop 2.** A system check preventing duplicate invoice recording within the same case (unless explicitly flagged as a correction) would eliminate the 336 Loop 2 cases with minimal process change.
- **Accept Loop 1 as normal.** Split deliveries are a legitimate business pattern with no measurable negative impact. No intervention is recommended.

Task 8 Effects of Cancellations and Deletions

Procedure

We identified four cancellation/deletion-related activities: Delete Purchase Order Item, Cancel Invoice Receipt, Cancel Goods Receipt, and Reactivate Purchase Order Item. For each we measured: (i) how many cases are affected, (ii) what activities were already performed before the event (over-processing), and (iii) the time elapsed between case start and the cancellation/deletion.

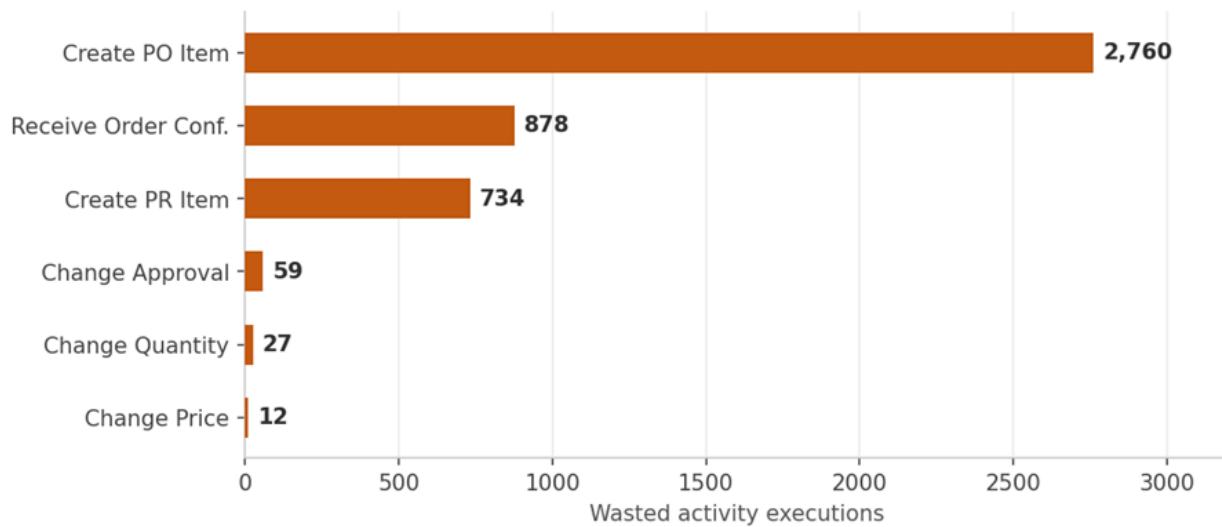
Summary Statistics

Delete PO Item 2,760 4.7% of all cases	Cancel Invoice Receipt 571 517 also have debit memo	Cancel Goods Receipt 44 rare but cascading	Median wasted time 3.2 days max 162 days
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Delete Purchase Order Item Over-Processing

In all 2,760 deletion cases, work had already been performed before the order was deleted. On average 1.6 activities were executed per case before deletion, yielding 4,470 wasted activity executions in total.

Figure 3 — Wasted activity executions prior to Delete PO Item

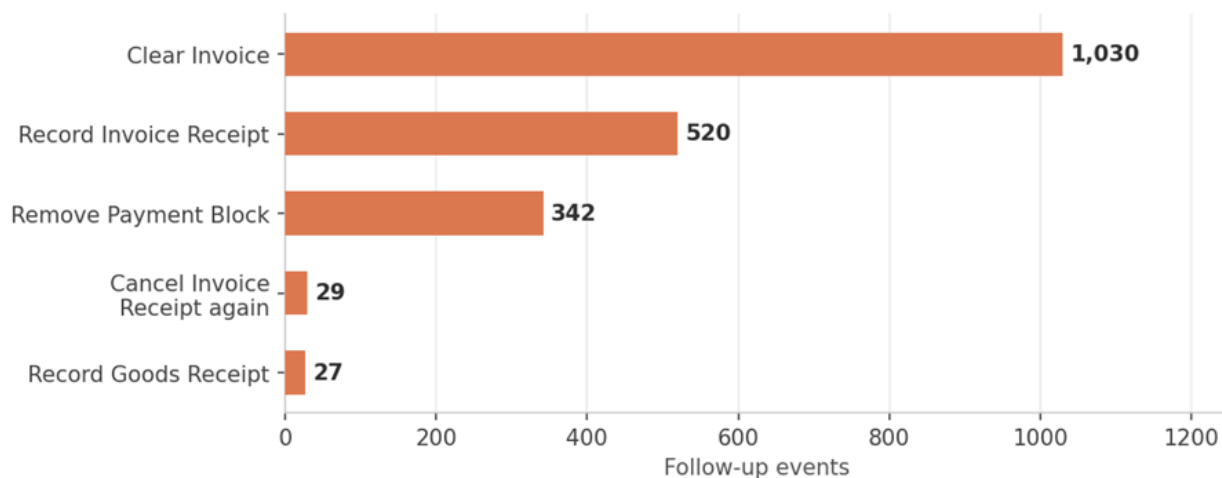


Median time from Create PO to Delete PO: 3.2 days. Mean: 6.8 days. The long tail (max 162 days) indicates some orders are carried for months before being formally deleted a sign of inadequate housekeeping.

Cancel Invoice Receipt Rework Cascade

Of 571 cases containing Cancel Invoice Receipt, 517 (90.5%) also include Vendor creates debit memo, indicating the primary cause is a vendor-side billing error requiring reversal. The chart below shows follow-up activities.

Figure 4 — Activities following Cancel Invoice Receipt



The dominant follow-up is Clear Invoice (1,030 events): cancel wrong invoice → vendor debit memo → process continues to closure. A secondary path (520 cases) shows a new Record Invoice Receipt, meaning the invoice is fully reissued.

Cancel Goods Receipt

Cancel Goods Receipt is rare (44 cases) but triggers a cascade: in 36 cases a Cancel Invoice Receipt follows, suggesting the goods return also invalidates the associated invoice, effectively restarting both phases.

Improvement Suggestions

Introduce pre-submission PO validation to reduce deletions. A mandatory checklist or approval step before PO creation would prevent the 4,470 wasted activity executions.

Work with vendors to improve invoice accuracy. Since 90.5% of Cancel Invoice Receipt cases are linked to vendor debit memos, vendor portal auto-validation against the PO before submission would significantly reduce this rework loop.

Implement a system-enforced timeout: POs inactive for more than 30 days should trigger an alert and prompt a proceed-or-delete decision, avoiding long-tail cases (up to 162 days).

Task 9 Effects of Change Activities

Procedure

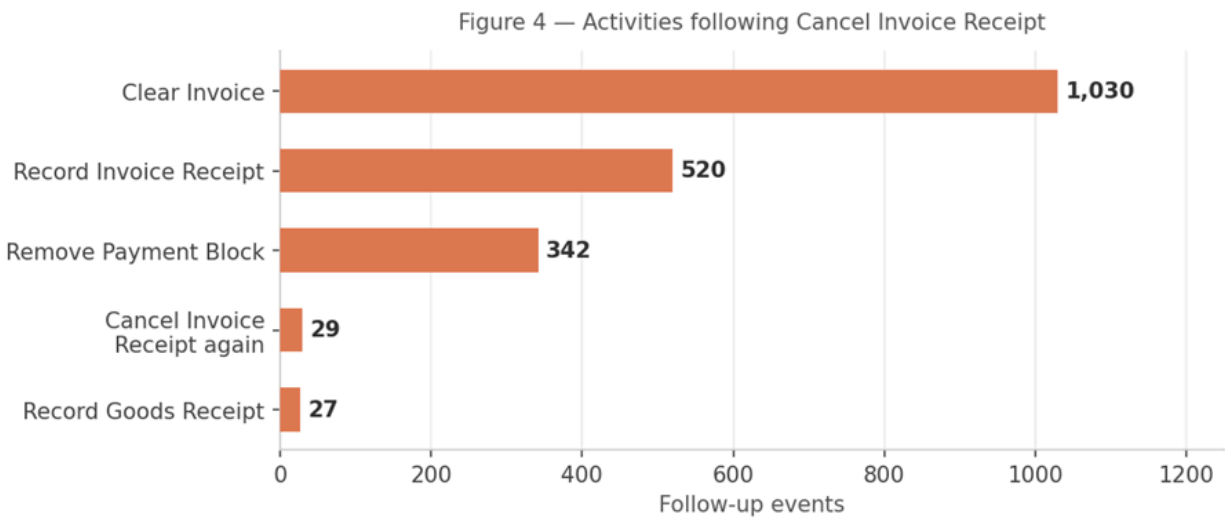
We identified five change activity types: Change Quantity, Change Price, Change Approval for Purchase Order, Change Delivery Indicator, and Change Storage Location. For each we computed the median cycle time of affected cases and compared it against a baseline (cases with no change activities: 79.2 days). We also examined which activities immediately follow each change event to understand downstream impact.

Summary Statistics

Baseline median CT 79.2 days no change activities	Change Quantity 1,992 +28 days vs baseline	Change Price 879 +19 days vs baseline	Change Approval 921 25.5% completed only
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Cycle Time Impact

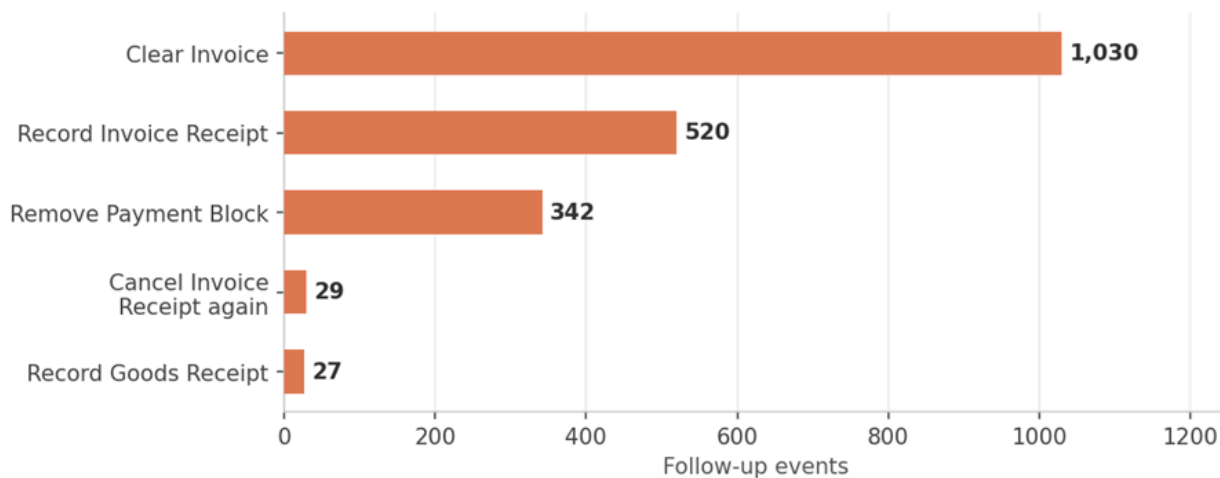
Cases with Change Quantity or Change Price show a substantial increase over the baseline. Change Approval cases have a misleadingly low overall median (4.7 days) because only 25.5% are completed the rest are mostly abandoned (211 end in Delete PO Item). For completed Change Approval cases the median rises to 93.4 days.



Downstream Effects

The charts below show which activities immediately follow Change Quantity and Change Price events respectively.

Figure 4 — Activities following Cancel Invoice Receipt



Findings

- **Change Quantity (+28 days):** In 95% of cases the change occurs before Goods Receipt, meaning order details were not finalised at creation. 83 cases loop back to another Change Quantity, indicating recurring instability.
- **Change Price (+19 days):** Price changes frequently trigger Remove Payment Block (174 events) the system blocks payment when the invoiced amount no longer matches the PO, requiring manual clearance.
- **Change Approval for Purchase Order:** Only 25.5% completion rate; 211 cases end in Delete PO Item, indicating this activity often precedes order abandonment rather than a routine approval update.
- **Change Delivery Indicator / Storage Location:** Rare (62 and 9 cases) and cycle times are near or below baseline administrative corrections with negligible process impact.

Improvement Suggestions

Improve PO data quality at creation. Since 95% of Change Quantity events occur before Goods Receipt, mandatory quantity confirmation with the vendor before PO submission would eliminate most corrections and recover the 28-day cycle time penalty.

Implement price tolerance rules for automatic payment block removal. When a price change is within an agreed tolerance (e.g., $\pm 2\%$), the block should lift automatically rather than requiring manual intervention, reducing the 174 Remove Payment Block events.

Monitor Change Approval cases proactively. Given that 46% end in Delete PO Item, a fast-track approval SLA (e.g., 48-hour limit) would either accelerate legitimate orders or surface cancellation decisions earlier, reducing wasted processing.

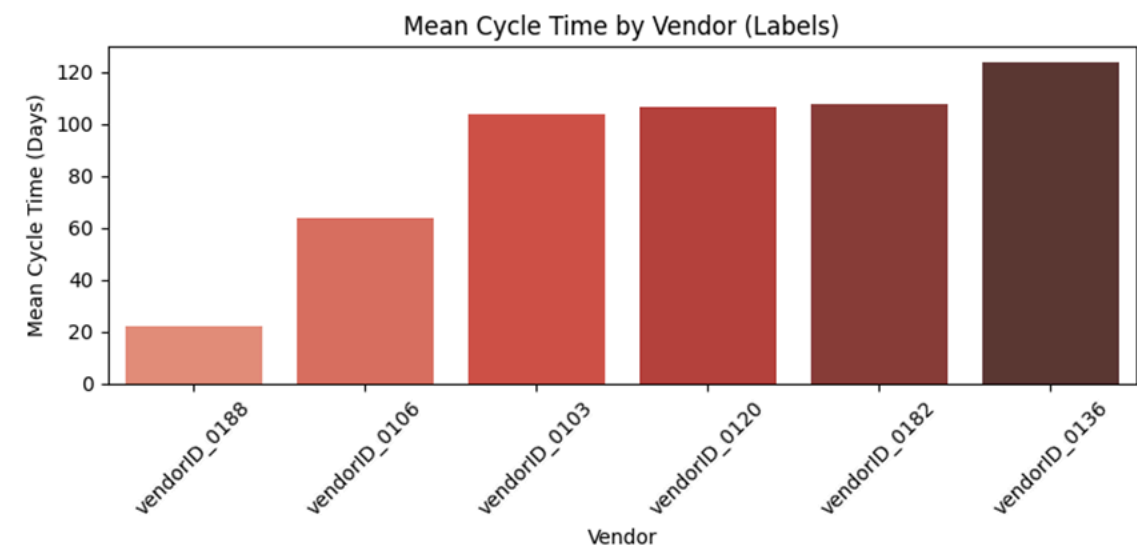
Task 10: Differences Between Vendors of Labels

Procedure

To rigorously assess vendor performance within the "Labels" sub spend area, we isolated all cases belonging to the 6 relevant vendors. We then decomposed the end-to-end cycle time into distinct operational phases by analysing directly-follows transition durations.

Specifically, we measured physical logistics performance (Delivery Time: 'Record Goods Receipt' to 'Record Invoice Receipt') versus administrative efficiency (Invoicing Lag and Payment Processing). Finally, we cross-referenced these durations with the occurrence rate of "Remove Payment Block" events to isolate whether delays were caused by supply chain friction or invoice compliance issues.

Mean Cycle Time by Vendor



Findings

- **Vendor 0188 (Consignment Model):** This vendor is the most efficient. Their process is the fastest, taking only 22.3 days on average, and they have the fewest process steps. This is because they use a consignment model, which avoids the usual manual invoice processing.
- **Vendor 0182 (Fast Delivery, Slow Admin):** This vendor has the fastest delivery time (9.3 days). However, they suffer from administrative lag, taking 13.5 days on average just to record their invoices into the system.
- **Vendor 0136 (Problematic):** This is the most problematic vendor, suffering from the longest overall cycle time (125.7 days). This is largely driven by severe payment delays. Over 60% of their cases require a "Remove Payment Block" event, indicating frequent invoice disputes.

Improvement Suggestions

- **Optimize Volume:** Shift more volume to Vendor 0188 to take advantage of the highly efficient consignment process.
- **Modernize Invoicing:** For Vendor 0182, implement digital invoicing (EDI) to completely eliminate the administrative recording delay.
- **Audit Contracts:** Restrict order volume to Vendor 0136 and conduct an audit to enforce Purchase Order (PO) compliance and reduce the number of invoice disputes.

Task 11: Free-Form Analysis: Case Granularity Comparison

Quick Overview

To answer the final free-form question, we used two different methods to uncover the hidden factors that cause delays in the purchasing process:

Part A: We merged individual line items into full purchasing documents and compared results. This revealed a hidden 25-day waiting penalty and 250,000 fake loops caused by parallel processing.

Part B: We trained a machine learning model (85% accurate) to predict cycle time, then used SHAP to rank which factors matter most. Top drivers: spend category, process complexity, Vendor 0136, order value, and document size.

Motivation

Throughout Tasks 4, 8, 9, and 10, we analysed the CoatXR P2P process at the line-item level, where each case corresponds to a single line item within a purchasing document. However, in practice, purchasing documents often contain multiple line items that are processed in parallel. This raises a critical question: does the choice of case granularity itself act as a hidden factor that affects our understanding of process performance? To investigate this, we pre-processed the data so that each case corresponds to a single purchasing document (merging all line items sharing the same Purchasing Document ID), and then re-examined cycle times and process model quality at both levels.

Methodology

We merged all 59,385 line-item-level cases into 9,118 document-level cases by grouping on the Purchasing Document column. For each document, the start time was set to the earliest event across all its line items and the end time to the latest. We then compared three dimensions across both granularities: (1) cycle time distributions, (2) the relationship between document size (number of line items) and cycle time, and (3) directly-follows loop counts as reported by the process model.

Finding 1, The Synchronization Penalty: A Hidden Factor Affecting Performance (+25 Days)

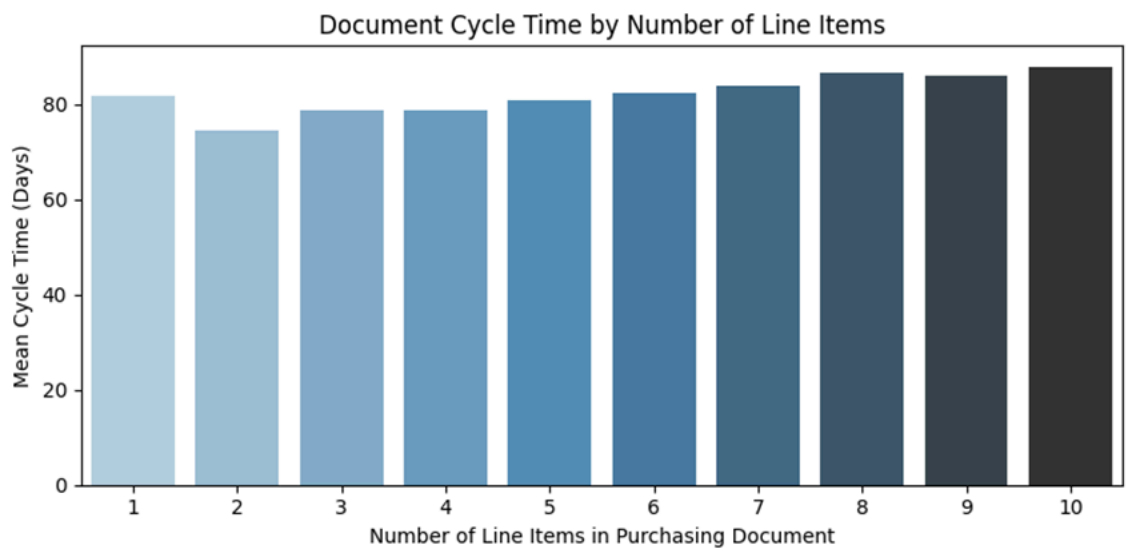


Figure: Cycle time increases steadily with the number of line items in a document, revealing the synchronization penalty.

At the line-item level, the mean cycle time is 74.8 days. At the document level, the mean cycle time jumps to 99.4 days, a difference of approximately 25 days. This gap is the synchronisation penalty: a purchasing document cannot be considered complete until its slowest line item finishes. Furthermore, cycle time increases steadily with the number of line items in a document,

documents with 5+ items take significantly longer than single-item documents. This reveals that document size is a previously unidentified factor affecting process performance: the more items bundled into one purchasing document, the longer the end-to-end completion time due to synchronisation waiting.

Finding 2, The Multi-Instance Parallelism Artifact: Process Model Distortion

At the line-item level, the process model reports 2,223 rework loops. At the document level, this figure explodes to 259,005 loops, a 116-fold increase. This is not genuine rework. Because multiple line items within a single document undergo the same activities (e.g., Record Goods Receipt, Vendor creates invoice) concurrently, the document-level process model misinterprets this parallelism as repeated executions of the same activity, creating the appearance of massive rework loops that do not exist in reality.

Impact on Earlier Analysis Tasks

This finding has direct implications for the reliability of our earlier analyses:

- **Task 4 (Deviations):** The ideal flow detection (39% compliance) would be impossible at the document level because fake parallelism loops would make virtually every case appear deviant.
- **Task 8 (Cancellations):** Measuring wasted effort per cancelled item (4,470 wasted executions) requires item-level precision. At the document level, cancelling one item does not close the document, so the wasted work would be hidden within the ongoing document.
- **Task 9 (Changes):** The +28 day cycle time impact of Change Quantity events would be masked at the document level by the 25-day synchronisation penalty, making it impossible to isolate the true effect of changes.
- **Task 10 (Vendors):** Vendor-specific transition durations (e.g., delivery speed of 9.3 days for Vendor 0182) would be diluted at the document level because a single document may contain items from different vendors or different spend areas.

Process Improvement Opportunities

- **Reduce document size:** Where possible, split large multi-item purchasing documents into smaller, independent orders so that each can complete without waiting for unrelated items. This could recover up to 25 days of hidden waiting time.
- **Implement staggered invoicing:** Rather than waiting for all items to arrive before processing invoices, allow partial invoicing per line item so that the financial close of each item is independent of the rest of the document.
- **Validate at item level first:** When performing process mining on multi-instance processes, always validate findings at the line-item level first. Document-level models produce misleading loop counts and inflated cycle times that can lead to incorrect improvement priorities.

Conclusion

The choice of case granularity is itself a significant, hidden factor affecting measured process performance. Document-level analysis inflates cycle times by 25 days due to synchronisation and produces 116 times more fake rework loops due to parallelism. This free-form analysis validates the analytical approach used throughout the project and reveals a concrete improvement opportunity: reducing purchasing document size to eliminate the synchronisation penalty.

Task 11 (Part B): ML + SHAP Analysis, What Factors Drive Cycle Time?

Motivation

The granularity analysis above revealed that case structure (document size) is a hidden factor affecting performance. But what are the other factors? To answer this systematically, we trained a machine learning model (XGBoost) to predict cycle time from case attributes and activity occurrence flags, then used SHAP (SHapley Additive exPlanations) to rank which factors have the largest impact on cycle time.

Methodology

We trained an XGBoost Regressor (300 trees, max_depth=6) on 46,229 completed cases using 51 features:

- **Categorical:** Vendor, Item Type, Sub spend area text, Spend area text, Document Type, Item Category, GR-Based Inv. Verif.
- **Numeric:** event_count, Cumulative net worth (EUR), doc_line_item_count.
- **Activity flags:** Binary indicators for whether each non-standard activity occurred (e.g., Remove Payment Block, Change Quantity, Change Price, Cancel Invoice Receipt).

Model Performance

The model achieved an R-squared of 0.8548, meaning it explains 85% of the variance in cycle time. The Mean Absolute Error is 9.55 days. This high accuracy confirms that cycle time is largely predictable from case attributes and activity patterns.

SHAP Feature Importance

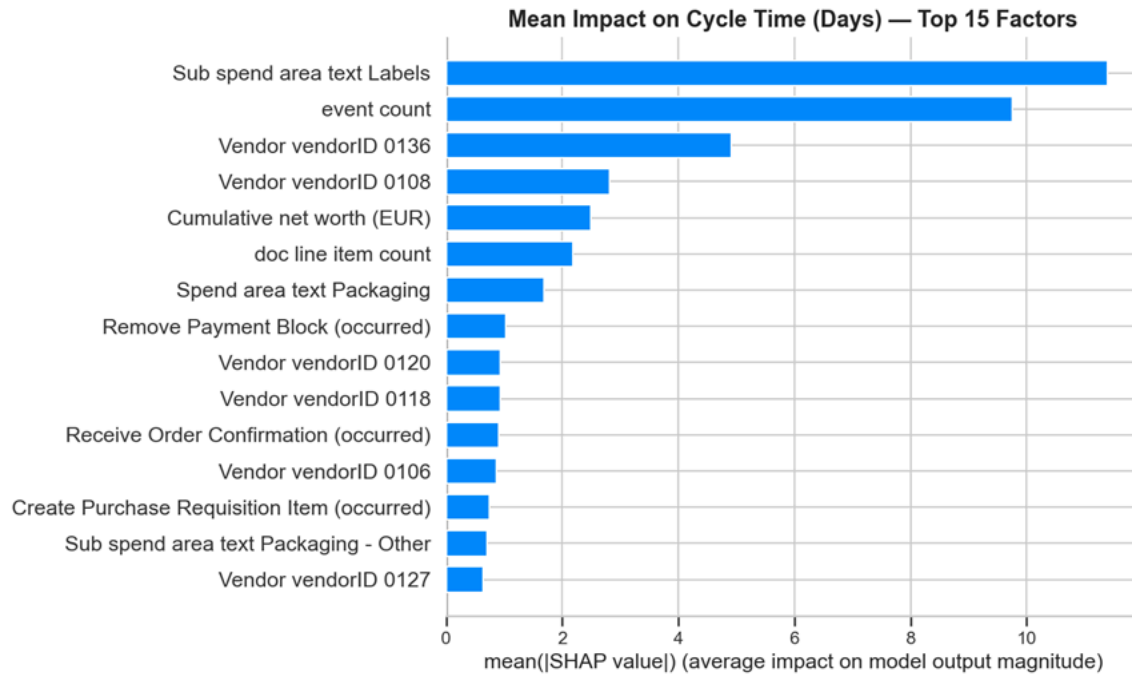


Figure: Mean absolute SHAP values showing the top 15 factors affecting cycle time.

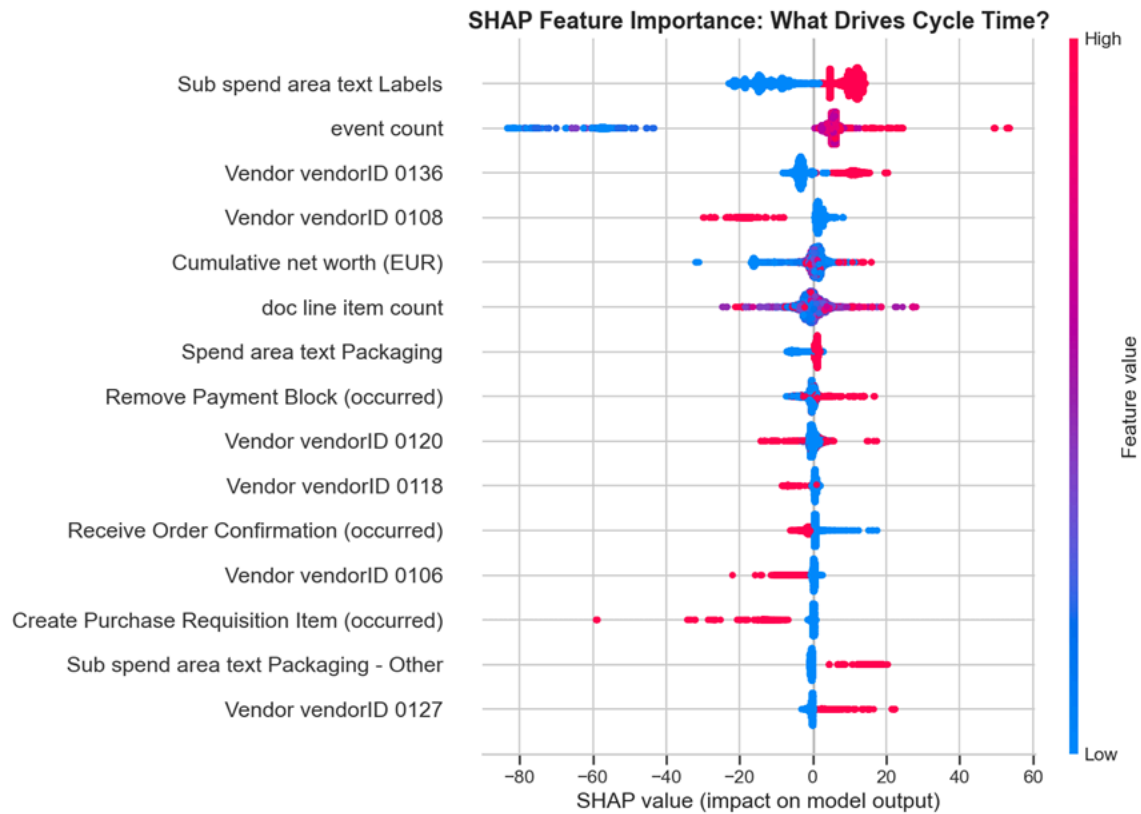


Figure: SHAP beeswarm plot. Each dot is a case. Red = high feature value, blue = low. Position on x-axis shows the impact on cycle time (days).

Key Findings

- **Sub spend area "Labels" (+11.4 days):** The single most impactful factor. Labels cases consistently take longer, confirming the vendor-specific delays identified in Task 10.
- **Event count (+9.8 days):** Cases with more activities (rework, changes, payment blocks) take significantly longer. This quantifies the cost of process complexity.
- **Vendor 0136 (+4.9 days):** This vendor systematically adds nearly 5 days to cycle time, confirming it as the worst performer (Task 10).
- **Cumulative net worth (+2.5 days):** Higher-value orders take longer, likely due to additional approval steps or more rigorous invoice verification.
- **Document line item count (+2.2 days):** Confirms the synchronisation penalty from Part A, more items in a document means longer cycle times.
- **Remove Payment Block (+1.0 days):** Each occurrence of this activity independently adds a day to cycle time, confirming findings from Tasks 4 and 9.

Data-Driven Process Improvement Priorities

Based on the SHAP ranking, the highest-impact improvements in order of priority are:

1. Reduce process complexity (event count) by automating invoice matching and eliminating unnecessary manual steps.
2. Audit and restrict Vendor 0136, they systematically add 5 days to every case they handle.
3. Implement tolerance-based automatic payment block removal to eliminate the 1-day penalty per occurrence.
4. Split large purchasing documents into smaller orders to reduce the synchronisation penalty.

Acknowledgement:

This project used AI language models as a support tool. Specifically, we took help with structuring and formatting the written report, debugging Python notebooks, and fixing code errors during analysis. All analytical decisions, findings, and conclusions are our own.